



Preparation and characterization of PVA/Chitosan cross-linked 3D scaffolds for liver tissue engineering

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ARTICLE INFO

Article history:

Available online 1 March 2023

Keywords:

Polyvinyl alcohol
Chitosan
3D bioprinting
Cross-linking
Liver tissue engineering

ABSTRACT

In this study, chitosan, and PVA are used to fabricate scaffolds using the 3D bioprinting technique for liver tissue engineering. PVA and chitosan hydrogels are cross-linked by the glutaraldehyde, genipin, and freeze-thaw methods. Hydrogels are then characterized by different techniques such as SEM, FTIR, swelling ratio, and biodegradability tests. Further, bioink is prepared using Chitosan/PVA hydrogel and HepG2 cells. Gels cross-linked by genipin were mechanically stable and also demonstrated higher cell viability compared to gels cross-linked by other methods. Overall, the chitosan-PVA cross-linked 3D hydrogels indicated substantial promise for liver tissue engineering.

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Selection and peer-review under responsibility of the scientific committee of the International e-Conference on Biopolymers.

1. Introduction

The liver is an organ that is crucial to preserve the body's equilibrium by involving in various metabolic processes. Liver diseases are one of the lethal diseases that would cause death. Although complete liver transplantation offers successful treatment options for individuals with end-stage liver disorders, the lack of organ donation restricts its applicability [1]. Liver tissue regeneration aims to create a completely functional organ that could be used to treat acute and chronic liver diseases. Liver tissue constructs generated *in vitro* are being used in organ transplantation, and have now been considered the gold standard treatment for liver disorders. Hepatocyte culture in three-dimensional (3-D) scaffolds has recently received attention as a promising method to aid in the regeneration of liver tissue. The 3-D culture method offers a state akin to the micro-environment to preserve hepatocyte activity without toxicity, in contrast to the traditional 2-D culture method. The choice of scaffolding materials is crucial for 3-D culture [2,3].

Chitosan is a linear amino hetero-polysaccharide that is produced by deacetylating chitin and has the properties of being highly biocompatible, biodegradable, low in toxicity, non-

antigenic, and absorbable. Additionally, the structure of chitosan and glycosaminoglycans, which make up the liver's extracellular matrix (ECM), are comparable [4,5]. Besides, chitosan in combination with other polymers has been frequently explored to ameliorate the physicochemical as well as biological properties. A synthetic, non-toxic, water-soluble, and biocompatible polymer with good characteristics is polyvinyl alcohol (PVA). PVA is considered a promising biomaterial owing to its good swelling qualities, biocompatibility, nontoxicity, and lack of carcinogens. Besides, PVA's water solubility necessitates cross-linking before hydrogel synthesis [6,7]. PVA/Chitosan scaffolds offer benefits of improved porosity, improved mechanical properties, and higher biocompatibility. Further, this blend mimics ECM composition, which is advantageous for cell division and proliferation.

In polymer chemistry, cross-linking is a stabilizing procedure that results in the multidimensional extension of the polymeric chain network structure. A cross-link is a bond covalent/ionic that connects two polymer chains. Cross-linking turns a liquid polymer into a "solid" or "gel" by limiting the segmental movement of the polymer chains. Ideally, by interconnecting the chains, a liquid polymer (where the chains are free to flow) can be transformed into a "solid" or "gel" [8]. Various studies have documented PVA-Chitosan hydrogels with various cross-linking techniques, such as multiple freeze-thaw cycles and adjusting the PVA's degree of hydrolysis [4]. However, the most used synthetic cross-linker, glu-

Abbreviations: CH, Chitosan; PVA, Poly vinyl alcohol; HepG2, Hepatoblastoma.

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taraldehyde, is toxic and significantly reduces the biocompatibility of cross-linked materials [8].

A natural crosslinker called genipin, the aglycon of the geniposide, is made from the fruit of the *Gardenia jasminoides* Ellis plant and has been exploited as a promising chemical fixative with negligible cytotoxicity [6,9]. Because of its lower cytotoxicity, higher cross-linking effectiveness, and improved biocompatibility, genipin has been widely employed to cross-link materials for biomedical purposes in recent years. Genipin may be a viable biomaterial cross-linker by lowering xenograft implantation in liver tissue engineering due to its considerable anti-inflammatory bioactivity [10].

The human hepatoblastoma cell line HepG2 has a stable phenotype and some endogenous and exogenous metabolic functions in comparison with primary hepatocytes. HepG2 cells could reproduce many features of normal hepatocytes and were less prone to dedifferentiation when these cells were grown on the scaffolds [11,12].

The purpose of this study was to develop a PVA and chitosan-based 3D-printed scaffold to utilize the benefits of each material for liver tissue engineering. The chitosan component offered excellent biological cues, whereas the PVA component increased the scaffold's physical toughness. The mechanical and biological effects of the two biomaterials, chitosan, and PVA, were evaluated using different weight fractions of each. The scaffolds with the optimal chitosan-to-PVA ratio were evaluated for homogeneous surfaces which could influence cellular behavior.

2. Materials and methodology

PVA (P15629) with a degree of hydrolysis of 98–100 mol% was purchased from Nice Chemicals, India. Chitosan (90 % deacetylation) (AB-B-003) was purchased from Alfatek Systems, India; lactic acid (41242) from SRL Chemicals, India; and Genipin (53409) from SRL Chemicals, India; Glutaraldehyde solution of 50 % (v/v) (TC558M) was purchased from Himedia Laboratories, India. The human hepatoblastoma cell line HepG2 was purchased from ATCC (American-type cell culture), USA. Dulbecco's modified Eagle's medium-high glucose (DMEM) (11965092) from ThermoFischer Scientific, USA, 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2H-tetrazolium bromide (MTT) (MB186) were obtained from Himedia Laboratories, India. All other chemicals used were of analytical grade. Deionized double distilled water was used throughout the experiment.

2.1. Preparation of PVA-Chitosan hydrogel

Chitosan solution (2 %) w/v was prepared by dissolving it in lactic acid solution (3 %) v/v via magnetic stirring. Polyvinyl alcohol solution (15 %) w/v was prepared by dissolving 15 g of PVA powder in 100 ml distilled water via magnetic stirring at 120 °C. The blend was prepared by mixing two solutions at different ratios from 1:9 to 9:1 of PVA and chitosan respectively and based on the non-flow behavior of hydrogels, the PVA/Chitosan blended hydrogels were optimized experimentally at a ratio of 5:5. The above two solutions were blended and cross-linked by different cycles (1, 3, 5 and 7) of 12 hrs of freeze-thaw process, genipin of concentration 0.025 %, 0.05 %, 0.075 % and 0.1 % (w/v) for 8 h and glutaraldehyde of concentration 0.1 %, 0.2 %, 0.3 %, 0.4 % and 0.5 % (v/v) for about 8 h.

2.2. Characterization of PVA-Chitosan hydrogel

2.2.1. FTIR analysis

The FTIR analysis was carried out by FTIR spectroscopy to determine the functional groups of blended hydrogels. Cross-linked

hydrogel and un-crosslinked hydrogel which served as control were characterized by attenuated reflectance Fourier transform spectroscopy (Bruker, Alpha II, USA). The infrared spectra of the samples were measured over a wavelength range of 4000–650 cm⁻¹.

2.2.2. Swelling ratio

The swelling characteristics of hydrogels are important and must be known for applications of hydrogels in different fields. The dry scaffolds are immersed in water for a definite time period at room temperature (30 °C) and the water uptake was calculated by measuring its initial weight and final weight after swelling at regular intervals. Also, the equilibrium water content (EWC) of the hydrogel samples was analyzed. The % swelling ratio and the EWC were calculated using the equation as described– [13].

$$SR (\%) = \frac{W_s - W_d}{W_d} * 100 \quad (1)$$

Here, w_s is the final weight of hydrogel at time (24 h) and w_d is the initial weight.

$$EWC = (W_{24h} - W_0) / W_{24h} * 100 \quad (2)$$

Here, W_{24h} is the weight of the gel at equilibrium and W_0 is the initial weight.

2.2.3. Determination of viscoelasticity

Viscoelasticity refers to the property of materials that can show elastic and viscous behavior. The viscosity is determined by the resistance to deformation at a known rate. The two-plate model is used for explaining the rheological parameters and viscosity is calculated by using an equation [10]

$$\vartheta = \frac{\tau}{\dot{\gamma}} \quad (3)$$

2.2.4. Determination of shear storage modulus and loss modulus

The elastic behavior of a material is explained by the storage modulus (G'), and the viscous behavior is explained by the loss modulus (G''). Chemical bonding in materials where $G' > G''$, is strong. Chemical bonds are weaker in materials where $G'' > G'$. The point at which $G'' = G'$ is known as the crossover point or gel point. When the material reaches this stage, gelation has caused it to change from a liquid state to a solid state. The measurement of overall flow resistance as a function of rotational frequency is known as complex viscosity. The following equation can be used to compute it [13]

$$\vartheta^* = \frac{G^*}{\omega} \quad (4)$$

2.2.5. Biodegradability test

Hydrogels were immersed in PBS solution and water and then periodically removed, dried, and weighed to examine degradation behavior. The degradation rate was determined by the equation [9]

$$Weightloss (\%) = \frac{W_i - W_r}{W_i} * 100 \quad (5)$$

where w_i is the dry weight of film before degradation rate and w_r is the dry weight of film after degradation rate.

2.2.6. Optimization of printing parameters

The PVA/Chitosan hybrid hydrogels have been cross-linked using different methods and are then printed using the optimized parameters. Among the factors that can affect print quality are pressure and temperature. These parameters can be altered to enhance the hydrogels' printing quality. The use of needles with

Table.1
Optimization of parameters for Printability.

Needle gauge size	Pressure (KPa)
22G	30 KPa, 40 KPa, 50 KPa, 60 KPa, 70 KPa, 80 KPa & 90 KPa
25G	30 KPa, 40 KPa, 50 KPa, 60 KPa, 70 KPa, 80 KPa & 90 KPa

varied gauge sizes and varying pressures was optimized for the hydrogels' printability (Table 1). The scaffolds were printed using Cellink Inkredible + Extrusion-based 3D Bioprinter (Cellink S-10002-001, Sweden). [14,15].

2.2.7. Biocompatibility assays

Hepatoblastoma G2 (HepG2) cells were grown in DMEM media with 10 % fetal bovine serum (FBS) at 37 °C in an incubator with 5 % carbon dioxide (CO₂). HepG2 cells were seeded onto the UV-sterilized, printed scaffolds, which were then incubated at 37 °C.

2.2.7.1. MTT assay. The cell viability in the scaffolds was measured with an MTT assay. HepG2 cells were cultured in the scaffolds for 1, 3, and 5 days. The MTT was added to a final concentration of 0.5 mg/mL and incubated for 4 h at 37°C and 5 % CO₂. The formazan precipitates in scaffolds were solubilized in DMSO and the absorbance was measured at 570 nm on a multimode microplate reader (Perkin Elmer (Ensign) HH34000000).

2.2.7.2. Live/Dead assay. The viability of the HepG2 cells exposed to the hydrogel environments was evaluated using a LIVE/DEAD viability/cytotoxicity kit (Thermo Fisher, U.S.A). Before cell seeding, the materials were sterilized using UV irradiation. Later, calcein-AM stock solution and ethidium homodimer-1 stock solution were added to the wells and incubated for 1 hr at 37°C. Calcein-AM, a cell-permeant dye is converted to a green-fluorescent calcein by viable cells, and ethidium homodimer-1 binds to nucleic acids of cells with damaged membranes to produce red fluorescence. Images were acquired via fluorescent microscopy (Nikon Eclipse-TE2000-U). The cell viability was estimated through quantification of the number of live cells over the total number of cells [5,16].

2.2.7.3. F-Actin staining. The culture media was aspirated from the cells and the cells were washed with 1X PBS. Then the cells were fixed with 4 % paraformaldehyde for 1 h at room temperature, washes were done with PBS followed by permeabilization with 0.5 % Triton X-100 for 30 min at room temperature. Blocking was done with 1 % BSA for 1 h at room temperature and the cells were incubated overnight at 4 °C. Washes were performed gently with PBS. Later stained with phalloidin stain (1:200) for 30 mins at room temperature and washed with PBS. The cells were stained with 1:1000 diluted DAPI solution and they were visualized under the fluorescence microscope (NikonTE2000U) with filters.

2.3. Statistical analysis

Data were presented as means ± standard deviations. Origin 2022b was used in statistical analysis. The two-way ANOVA test was performed to determine statistically significant differences between the values of several experimental groups, and a p-value of 0.05 was considered statistically significant.

3. Results and discussion

3.1. Preparation and Optimization of PVA/Chitosan hydrogel

PVA solutions at 5 % (w/v) and 10 % (w/v) were highly watery and therefore unusable whereas 15 % (w/v) PVA solution demon-

strated good consistency and therefore, a 15 % PVA concentration was optimal and was used in the experiment. Chitosan gel was optimized at 2 % (w/v) of chitosan dissolved in 3 % (v/v) lactic acid. The PVA/Chitosan blended hydrogel was optimized at a ratio of 5:5 based on the non-flow behavior of hydrogels and cross-linking. The hydrogel was optimized with the following characteristics: 15 % PVA blended with 2 % chitosan (solvent system – 3 % lactic acid) in a ratio of 5:5. The vial inversion method was used to determine the gelation. The gelation was observed when the hydrogel stopped flowing and was recorded at room temperature. The gelation was observed by inverting the vials [11].

3.2. Cross-linking methods

On cross-linking by freeze–thaw method, the gels at 1st cycle of freeze–thaw showed non-flow behavior. The gels cross-linked with 0.025 % genipin and 0.4 % glutaraldehyde, exhibited non-flow behavior (Fig. 1).

3.3. Characterization of hydrogel

3.3.1. FTIR analysis

In Fig. 2. The O–H stretching is represented by the peak at 3276 cm⁻¹ and every hydrogel shows this peak. This is related to the vibrations of the N–H and –OH groups of chitosan, as well as the hydroxyl groups of PVA and the peak at 1081 cm⁻¹, which is related to the overlapping C–O stretching vibrations of chitosan and PVA [17]. Peaks 1580 cm⁻¹ and 1505 cm⁻¹ in the PVA/Chitosan mixture cross-linked by genipin indicate the existence of amine and the N–O bond, respectively. Identical peaks are observed in the PVA/Chitosan mixture that was cross-linked by glutaraldehyde by the peak at 1651 cm⁻¹ indicating the presence of the imine group. This peak also proves the Schiff base formation after cross-linking. [9].

3.3.2. Swelling ratio

The swelling ratio of hydrogels is affected by the different densities of cross-links. The lower the swelling ratio, the higher the cross-linking density. From Fig. 3 we can observe that glutaraldehyde and freeze–thaw cross-linking have a lower swelling ratio as compared to genipin cross-linking. Since there are minute differences, we can conclude that cross-linking density of all three methods is high. The PVA/Chitosan blend shows instability as the hydrogel dissolves in water very early.

The cross-linked film showed swelling and had strong water absorption capabilities Table 2. According to a study, the cross-linked films reached saturation, suggesting that cross-linking caused the structure of the films to become dense and stable and to induce dispersed swelling without changing the morphology of the film. The impact of cross-linking is responsible for the variation in swelling between the cross-linked and un-crosslinked films. [6].

3.3.3. Determination of viscoelasticity

As the shear rate increases, we observe that viscosity decreases in Fig. 5. This implies that the material is shear thinning. The apparent viscosity of PVA/CH hydrogel (uncross-linked) at a shear rate of 50 s⁻¹ is seen to be 234 mPa-s whereas, the apparent viscosity of PVA/CH hydrogels cross-linked by the freeze–thaw method, cross-linked by genipin, and cross-linked by glutaraldehyde are 698.68 mPa-s, 3895.1 mPa-s, and 1028.9 mPa-s respectively. However, at the same shear rate, gels with higher cross-linking indicated higher apparent viscosity. Further, the solutions' viscosities were evaluated using a viscometer. The PVA solution had a viscosity of 234 mPa-s, while the PVA/chitosan cross-linked by genipin had a viscosity of 3895.1 mPa-s. This is consistent with the PVA

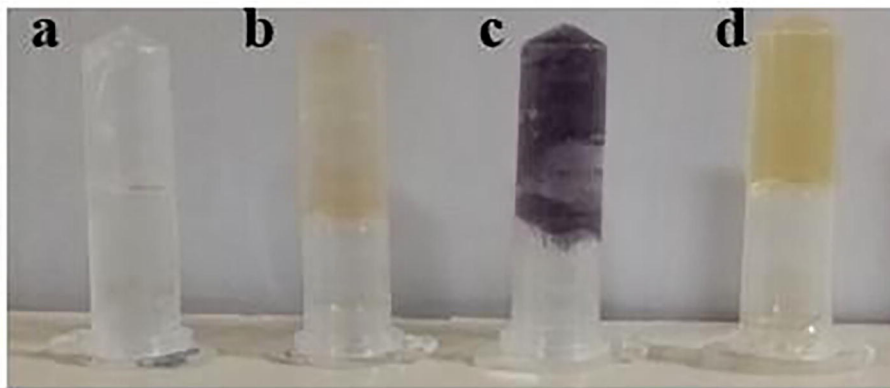


Fig. 1. A) PVA/Chitosan uncross-linked hydrogel; PVA/Chitosan hydrogel cross-linked by b) freeze–thaw method c) genipin & d) glutaraldehyde.

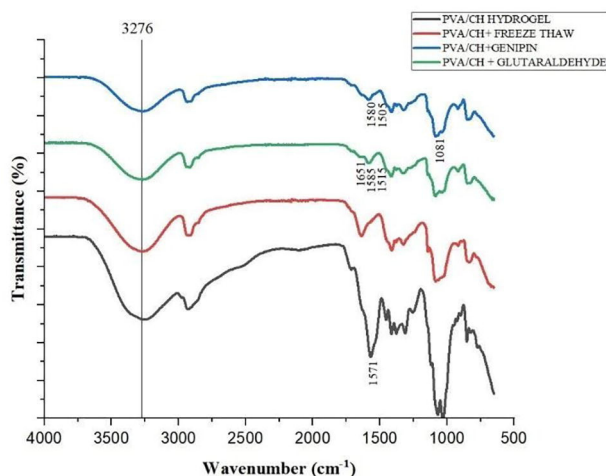


Fig. 2. FTIR spectrum of PVA/Chitosan un-cross-linked and cross-linked hydrogels.

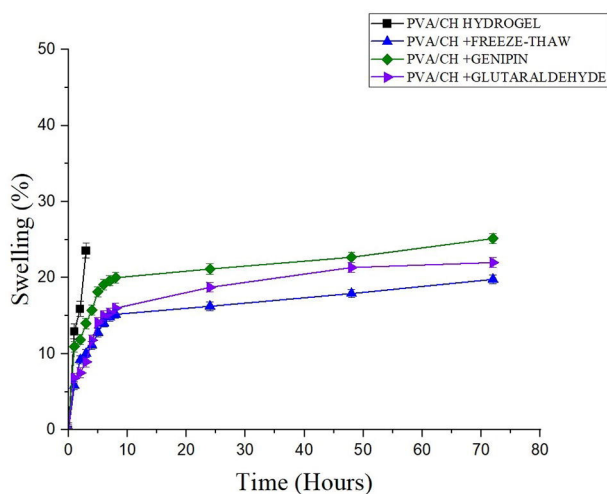


Fig. 3. Swelling ratio of un-crosslinked and cross-linked PVA-chitosan hydrogels ($n = 3$).

solution's linear increase in viscosity following its blending with various cross-linkers. High viscosity reduces the effects of surface tension while increasing the contact between two polymers, this effect is attributed primarily to hydrogen bonding. [18].

Table 2

Equilibrium Water Content of Hydrogels.

PVA/CH Hydrogels	EWC
Uncross-linked	100
Cross-linked by Freeze-Thaw	90.95
Cross-linked by Genipin	72.43
Cross-linked by Glutaraldehyde	88.05

The fluid's flow behavior is influenced by the shear rate and shear stress. Using the Herschel-Bulkley fluid model, which represents apparent strength and yield strength, respectively, and shear rate, one may explain the viscous flowing behavior. The apparent fluid viscosity at a shear rate of 1 s^{-1} is the consistency index. The ability of a material to thicken ($p > 1$), thin ($p < 1$), or maintain Newtonian behavior ($p = 1$) is expressed by the power law index or p . [19].

In Fig. 4. The shear rate increases in direct proportion to shear tension. As a result, we could affirm that the material is shear thinning. The parameter p is determined using the Herschel-Bulkley regression model to be 0.6429 for PVA/CH hydrogel, parameter p of 0.67851 for PVA/CH cross-linked by freeze–thaw method, parameter p of 0.9249 for PVA/CH cross-linked by genipin and the parameter p of 0.66102 for PVA/CH cross-linked by glutaraldehyde. The results indicate the shear thinning effect on both un-crosslinked and cross-linked gel.

3.3.4. Determination of storage and loss modulus

Fig. 5 indicates that complicated viscosity exhibits significant changes as angular frequency increases. Additionally, the complex viscosity exhibits slight increases with increasing angular frequency. This might be caused by a dense concentration of suspended solid particles.

It is noted that for PVA/CH hydrogel graph, $G'' > G'$. This suggests that the material's viscous behavior predominates over its elastic behavior. Bonds start to move more readily as a result of the material becoming rubbery. As a result, the material has lower mechanical strength. Here, it is seen that as angular frequency rises, complex viscosity slightly decreases. As a result, the material is shear thinning. Similarly, for PVA/CH cross-linked by the freeze–thaw method, and PVA/CH cross-linked by glutaraldehyde, $G'' > G'$ is noticed. The viscous behavior of the material dominates over its elastic behavior. The mechanical strength of the material is reduced, it is rubbery and the cross-linking density is lower. It has been found that as angular frequency increases, complex viscosity decreases. Thus, it is proven that the material is shear thinning. Whereas in the case of PVA/CH cross-linked by genipin, it is observed that $G' > G''$. This suggests that the material exhibits a

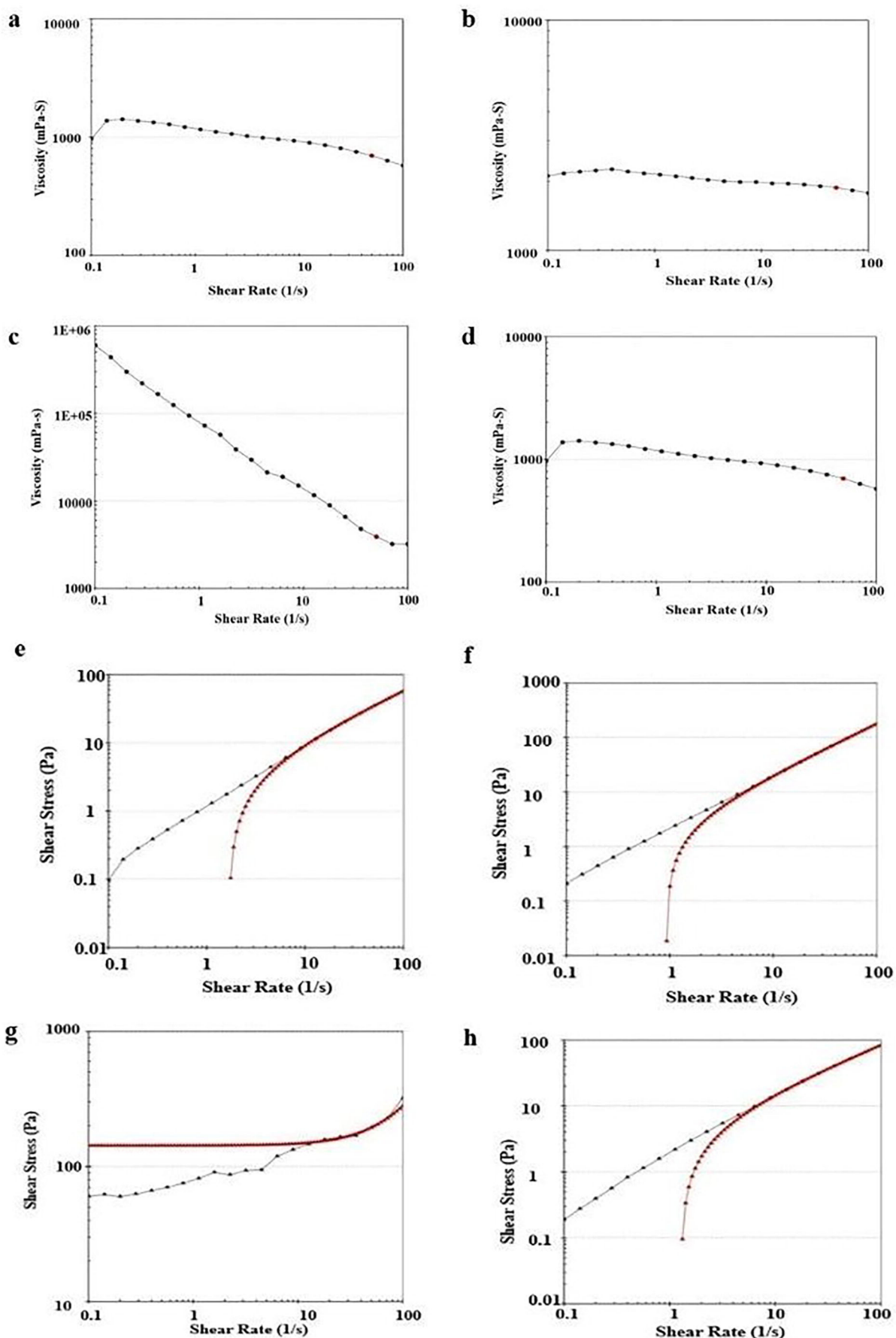


Fig. 4. Figure in the top row represents viscosity v/s shear rate v/s viscosity of PVA/CH hydrogel a) uncross-linked b) cross-linked by freeze-thaw method c) cross-linked by genipin d) cross-linked by glutaraldehyde and the bottom row represents the shear stress v/s shear rate of PVA/CH hydrogel e) uncross-linked f) cross-linked by freeze-thaw method g) cross-linked by genipin h) cross-linked by glutaraldehyde.

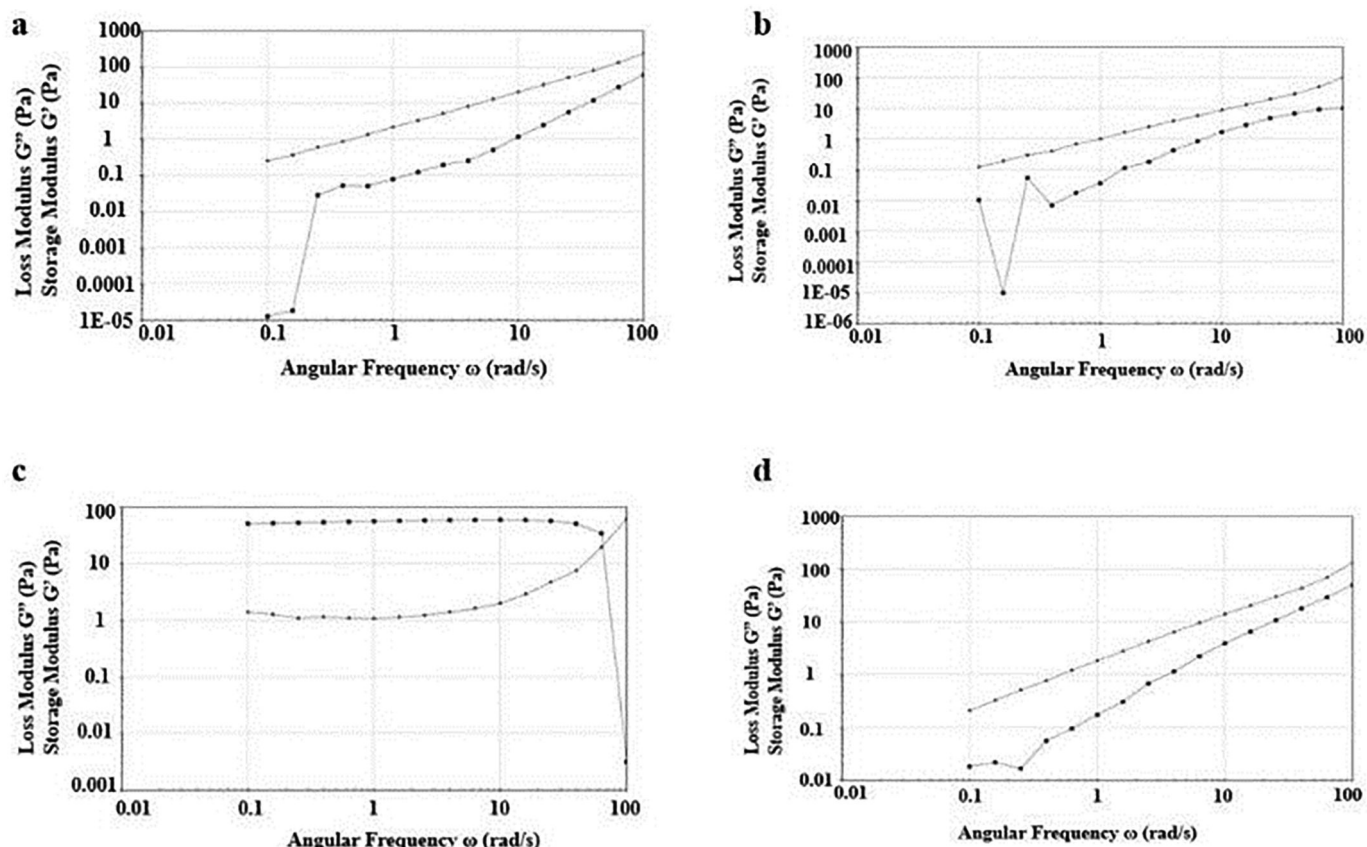


Fig. 5. The figure represents the G' and G'' v/s angular frequency of a) PVA/CH hydrogel b) PVA/CH cross-linked by freeze-thaw method c) PVA/CH cross-linked by genipin d) PVA/CH cross-linked by glutaraldehyde.

more significant elastic behavior than viscous behavior. As the angular frequency rises, it is seen that the storage modulus is nearly constant at a high value. This suggests that the material is strong and rigid mechanically. This demonstrates the high level of cross-linking [20].

3.3.5. Biodegradability test

Fig. 6a displays the hydrogels' percentage weight loss over a 7-day period in PBS. On day 3, the PVA/Chitosan blend hydrogel can be seen to have lost 100 % of its weight. This is brought on by the

hydrogel's lack of cross-linking. When compared to genipin and glutaraldehyde, which exhibit little variation in weight loss throughout the course of the experiment, PVA/Chitosan blended hydrogel cross-linked by freeze-thaw technique exhibits substantial weight loss. Cross-linked hydrogels can be inferred to have a low rate of biodegradability and to be stable over time as a result.

Fig. 6b displays the hydrogels' weight loss percentage throughout a 7-day period in water. On day 4, the PVA/Chitosan blend hydrogel without cross-linking exhibited 100 % weight loss. This variation reveals the hydrogel's instability. Both freeze-thaw and

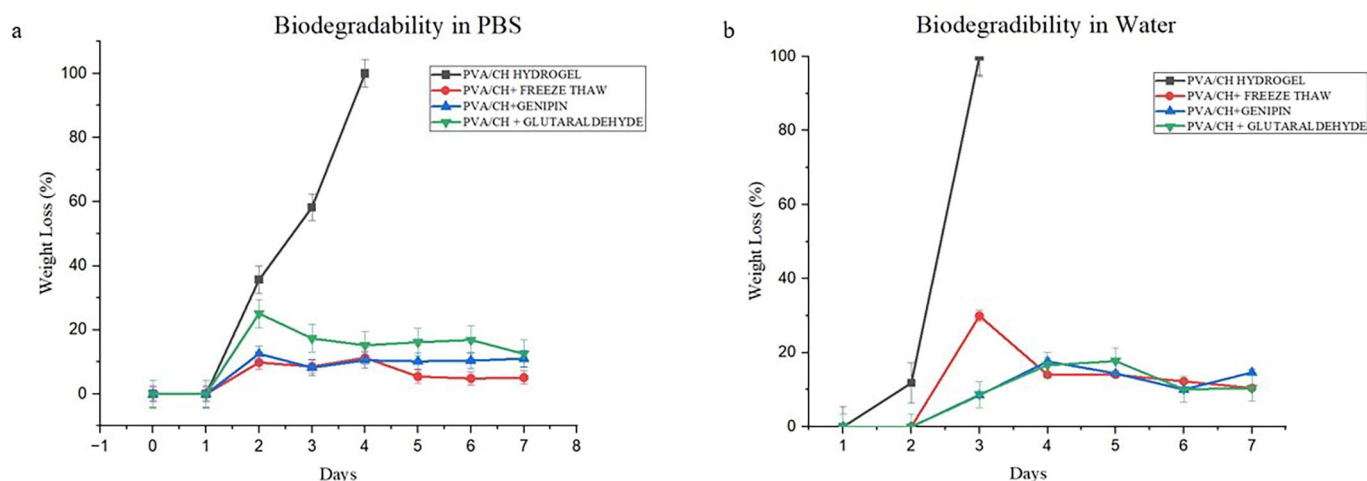


Fig. 6. Biodegradability of scaffolds in a) PBS and b) water ($n = 3$).

genipin cross-linking of PVA/Chitosan blended hydrogels result in close to identical rates of weight loss. The percentage loss is smaller in comparison to glutaraldehyde hydrogels, and as a result, these hydrogels degrade at lower rates than uncross-linked hydrogels.

According to the literature, [4,21] since the chains are joined by potent covalent bonds during cross-linking, the substance becomes insoluble. Although cross-linked materials can absorb solvents, they cannot dissolve in them whereas the uncross-linked gels degrade faster as there is no external covalent bond formation occurs due to a lack of cross-linking.

3.4. Optimization of printing parameters

The experimental data supplied shows how different printing parameters affect the results. In order to improve printing accuracy, write speed increases at higher print pressures have a bigger effect on strand width. In Fig. 7, a 22 G needle at a 37 °C temperature was used. As can be seen from Fig. 8, freeze-thaw scaffolds exhibit good shape integrity at pressures of 40 KPa. The printed glutaraldehyde scaffolds at various pressures did not exhibit any better shape fidelity and the genipin scaffolds at a pressure of 80 KPa exhibited good shape integrity [22].

At a temperature of 37 °C, a 25 G needle was used in Fig. 8. From the images, it can be shown that hydrogel is printable and exhibits better shape integrity for freeze-thaw scaffolds at a pressure of 50 KPa, for genipin scaffolds at a pressure of 80 KPa, and for glutaraldehyde scaffolds at a pressure of 60 KPa [14,23].

According to Georgia et al, high-pressure extrusion of viscous liquids yields high-fidelity printed scaffolds. However, at lesser concentrations, the inks are unable to generate consistently extruded filament, resulting in a dot structure rather than a uniform grid [24].

3.5. Biocompatibility assay

3.5.1. MTT assay

Fig. 8 highlights that the genipin cross-linked PVA/Chitosan gels have the highest level of cell viability. This might be because genipin has low cytotoxicity [10] and is a natural cross-linking agent. PVA/Chitosan blended hydrogel that has been cross-linked using the freeze-thaw process also exhibits good cell viability, this might be because the mechanism of cross-linking is physical and cytotoxicity is decreased since no chemical agents are used. Low cell viability is present in the glutaraldehyde cross-linked hydrogel, this might be because it acts as a chemical cross-linking agent and induces cytotoxicity. The freeze-thaw and genipin cross-linked hydrogels exhibit greater cell viability in comparison with uncross-linked hydrogels and TCPS (Tissue Culture Polystyrene) control. [9].

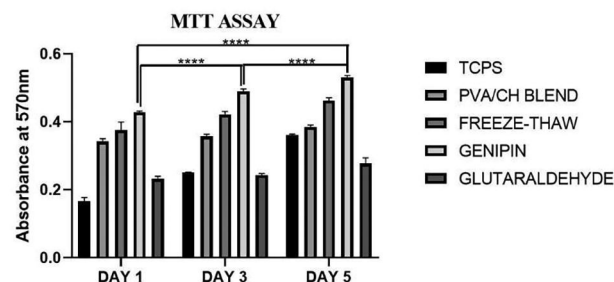


Fig. 8. MTT Assay of HepG2 cells seeded on printed scaffolds after 1, 3, and 5 days of cell culture. ****p < 0.0001, (n = 3).

3.5.2. Live/Dead assay

The control, TCPS, in Fig. 9 displays more live cells than dead cells. Because there are more cells evident, the PVA/Chitosan blended scaffold cross-linked by genipin has minimal cytotoxicity. Also, there are more live cells than dead ones. Thus, it can be inferred that genipin cross-linking is better than other cross-linking methods for improving cell survival and proliferation on the scaffold. [5].

3.5.3. F-Actin staining

In Fig. 9 and supplementary figure S1, genipin cross-linked scaffolds had good morphology and a high proliferation rate when compared to other cross-linked and uncross-linked scaffolds. Compared to glutaraldehyde scaffolds (Fig. 9j), genipin cross-linked scaffolds (Fig. 9i) were more effective in promoting HepG2 cell growth and maintaining cell viability similar to other literature [10] Although genipin produced from *Gardenia jasminoides* has been demonstrated to have less cytotoxicity than glutaraldehyde, synthetic cross-linkers often show cytotoxicity. This study [10] showed that genipin cross-linked chitosan/gelatin scaffolds could offer larger pores for the exchange of oxygen and nutrients than glutaraldehyde cross-linked scaffolds. In the biocompatibility assay, HepG2 cells have grown uniformly on chitosan/gelatin scaffolds that have been cross-linked with genipin.

4. Conclusion

In this work, 3D hydrogel scaffolds were fabricated using a blend of PVA and Chitosan. These hydrogels were cross-linked by physical and chemical methods. Further, the hydrogels were characterized using different methods. The presence of amine groups in the PVA/Chitosan blend cross-linked by genipin and glutaraldehyde makes the structures of hydrogel cationic in nature. This can make the scaffolds interact with the negatively charged cell surface easily and thus, makes the scaffold bio-adhesive. In contrast to glutaraldehyde, genipin was able to combine with free amino groups to create a more stable tertiary amine structure than



Fig. 7. Printed scaffolds of the freeze-thaw cross-linked hydrogel at pressure a) 40 KPa and 22G needle b) 70 KPa and 22G needle. c) 50 KPa and 25G needle d) 90 KPa and 25G needle and e) 70 KPa and 25G needle.

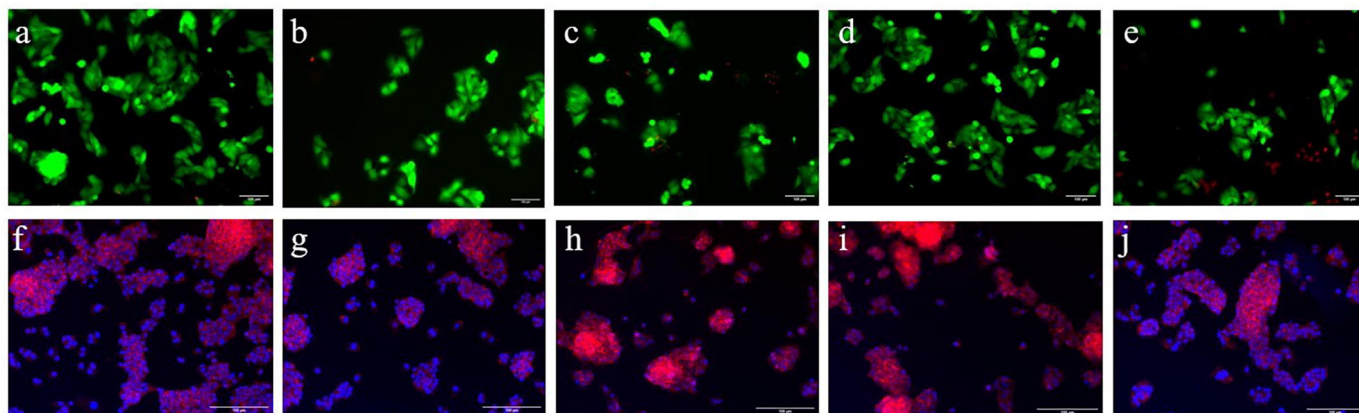


Fig. 9. Figure in the top row represents Live/Dead Assay of HepG2 Cells on a) TCPS b) PVA/Chitosan hydrogel scaffold c) PVA/Chitosan cross-linked by freeze-thaw d) PVA/Chitosan cross-linked by genipin e) PVA/Chitosan cross-linked by glutaraldehyde. Scale bar 100 μm . The bottom row represents F-Actin staining HepG2 cells on f) TCPS g) PVA/Chitosan hydrogel scaffold h) PVA/Chitosan cross-linked by freeze-thaw i) PVA/Chitosan cross-linked by genipin j) PVA/Chitosan cross-linked by glutaraldehyde. Scale bar 100 μm .

the Schiff base. The synthetic chemical cross-linkers may have cytotoxic effects, but genipin since it was extracted from *Gardenia jasminoides* Ellis showed lesser cytotoxicity than glutaraldehyde. Viscosity played a crucial role in printability. If the bioink has low viscosity, the structure may deform easily and if it has a very high viscosity, it may not extrude from the nozzle. The storage and loss moduli provide insight into the mechanical strength and degree of cross-linking. Materials with storage modulus greater than loss modulus show better mechanical strength and a higher degree of cross-linking. Materials with greater loss modulus show viscous behavior, are rubbery, have low mechanical strength, and low degree of cross-linking. Our study indicated that genipin cross-linked gels facilitated uniform HepG2 cell growth. Besides, the ability of genipin cross-linked scaffolds to encourage the adhesion, in-growth, and proliferation of HepG2 cells has been supported using a live/dead assay and F-actin staining. HepG2 cells could sustain normal cell adherence and proliferation when grown in the genipin cross-linked PVA/Chitosan 3-D microenvironment. The structural properties, cell survival, and proliferation of PVA/Chitosan scaffolds cross-linked with various cross-linkers were all compared. Overall, genipin cross-linked PVA/Chitosan scaffolds could be a promising 3-D cell culture system for liver tissue engineering applications.

CRediT authorship contribution statement

S. Supriya Bhatt: Writing – original draft, Methodology, Software. **Goutam Thakur:** Supervision, Writing – review & editing, Formal analysis, Validation. **Manasa Nune:** Conceptualization, Visualization, Investigation, Data curation, Project administration, Funding acquisition, Resources.

Data availability

Data will be made available on request.

Declaration of Competing Interest

Authors declare that there are no competing interests.

Acknowledgments

The authors would like to acknowledge the Department of Science & Technology, India for the SERB POWER grant

(SPG/2021/003703) and SERB Start-up research grant (SRG/2019/002130) for the financial support and the Manipal Institute of Regenerative Medicine for the infrastructural support. We would also like to thank Dr. T.M.A. Pai Ph.D. Scholarship from Manipal Academy of Higher Education. We acknowledge Ms. Aishwarya Bose, MIT Biomedical engineering B. Tech intern for her contribution to the present work. We also would like to thank Prof. Jyothi Prasanna, MIRM for providing us with the HepG2 cells.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.matpr.2023.02.251>.

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